# THE MOR FUNCTOR FROM A FINITE FREE AFFINE SCHEME INTO AN AFFINE SCHEME IS REPRESENTABLE BY AN AFFINE SCHEME

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*Editorial note.* Statement span is the whole lemma environment lines 20099-20105; proof span is lines 20107-20168.

*Editorial note.* The proof ends with the sentence 'Some details omitted.'; that closing sentence is retained verbatim. Under the project rule the main derivation must be complete and only routine verification may remain, which is the case here: the omitted part is the mechanical check that the assigned map P/J -> R' is well-defined and that the resulting T -> W has the required pullback property, and every ingredient for that check is already constructed in the retained text.

*Editorial note.* The proof cites no label and no bibliography entry, so no external_references are needed.

*Editorial note.* The original writes the map name as b_{univ} inside a math display and as b_univ in surrounding prose; both are retained verbatim.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The functor Mor_S(Z,X) of S-morphisms, finite free modules and bases, presentations Gamma(X,O_X) = R[x_i]/(f_k), the Yoneda lemma, and pushforward/pullback of algebras along T -> S. The representing-object construction is retained in full.

**Lemma 0.1.** *Let $Z \to S$ and $X \to S$ be morphisms of affine schemes. Assume $\Gamma(Z, \mathcal{O}_Z)$ is a finite free $\Gamma(S, \mathcal{O}_S)$-module. Then $Mor_S(Z, X)$ is representable by an affine scheme over $S$.*

*Proof.* Write $S = \mathrm{Spec}(R)$. Choose a basis $\{e_1, \ldots, e_m\}$ for $\Gamma(Z, \mathcal{O}_Z)$ over $R$. Choose a presentation

$$\Gamma(X, \mathcal{O}_X) = R[\{x_i\}_{i \in I}]/(\{f_k\}_{k \in K}).$$

We will denote $\overline{x}_i$ the image of $x_i$ in this quotient. Write

$$P = R[\{a_{ij}\}_{i \in I, 1 \leq j \leq m}].$$

Consider the $R$-algebra map

$$\Psi : R[\{x_i\}_{i \in I}] \longrightarrow P \otimes_R \Gamma(Z, \mathcal{O}_Z), \quad x_i \longmapsto \sum\nolimits_j a_{ij} \otimes e_j.$$

Write $\Psi(f_k) = \sum c_{kj} \otimes e_j$ with $c_{kj} \in P$. Finally, denote $J \subset P$ the ideal generated by the elements $c_{kj}$, $k \in K$, $1 \leq j \leq m$. We claim that $W = \mathrm{Spec}(P/J)$ represents the functor $Mor_S(Z, X)$.

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

First, note that by construction $P/J$ is an $R$-algebra, hence a morphism $W \to S$. Second, by construction the map $\Psi$ factors through $\Gamma(X, \mathcal{O}_X)$, hence we obtain an $P/J$-algebra homomorphism

$$P/J \otimes_R \Gamma(X, \mathcal{O}_X) \longrightarrow P/J \otimes_R \Gamma(Z, \mathcal{O}_Z)$$

which determines a morphism $b_{univ} : W \times_S Z \to W \times_S X$. By the Yoneda lemma $b_{univ}$ determines a transformation of functors $W \to Mor_S(Z, X)$ which we claim is an isomorphism. To show that it is an isomorphism it suffices to show that it induces a bijection of sets $W(T) \to Mor_S(Z, X)(T)$ over any affine scheme $T$.

Suppose $T = \mathrm{Spec}(R')$ is an affine scheme over $S$ and $b \in Mor_S(Z, X)(T)$. The structure morphism $T \to S$ defines an $R$-algebra structure on $R'$ and $b$ defines an $R'$-algebra map

$$b^\sharp : R' \otimes_R \Gamma(X, \mathcal{O}_X) \longrightarrow R' \otimes_R \Gamma(Z, \mathcal{O}_Z).$$

In particular we can write $b^\sharp(1 \otimes \overline{x}_i) = \sum \alpha_{ij} \otimes e_j$ for some $\alpha_{ij} \in R'$. This corresponds to an $R$-algebra map $P \to R'$ determined by the rule $a_{ij} \mapsto \alpha_{ij}$. This map factors through the quotient $P/J$ by the construction of the ideal $J$ to give a map $P/J \to R'$. This in turn corresponds to a morphism $T \to W$ such that $b$ is the pullback of $b_{univ}$. Some details omitted. $\square$